The Precision-Recall Trade-off (Optional) ⚖️

Ideally, we want a model to have both high precision and high recall. However, in practice, these two metrics are often in conflict: improving one can hurt the other. This relationship is managed by adjusting the model's **classification threshold**.

The Role of the Threshold

Recall that logistic regression outputs a probability $f(x)$ (a number between 0 and 1). We typically use a **threshold of 0.5** to make a prediction:

- If $f(x) \geq 0.5$, predict $\hat{y} = 1$
- If $f(x) < 0.5$, predict $\hat{y} = 0$

We can *change* this threshold to prioritize either precision or recall, depending on our goal.

Two Scenarios for Adjusting the Threshold

Scenario 1: Prioritizing High Precision (Being "Confident")

- **Goal:** We want to be very sure that when the model predicts $\hat{y} = 1$, it is correct. We want to avoid **False Positives (FP)**.
- **Example:** Predict $\hat{y} = 1$ (has disease) only if a treatment is very expensive, invasive, or risky. We don't want to treat healthy people.
- **Action: Raise the threshold** (e.g., to 0.7 or 0.9).
- **Result:**
  - **Higher Precision:** The model only predicts `1` when it's highly confident, so it makes fewer False Positives.
  - **Lower Recall:** The model will be too cautious and miss more of the actual positive cases (it will have more **False Negatives, FN**).

Scenario 2: Prioritizing High Recall (Being "Safe")

- **Goal:** We want to find *all* possible positive cases. We want to avoid **False Negatives (FN)**.
- **Example:** Predict $\hat{y} = 1$ (has disease) if a disease is very dangerous if *not* treated, but the treatment itself is cheap and safe. We'd rather treat 10 healthy people than miss one sick person.
- **Action: Lower the threshold** (e.g., to 0.3).
- **Result:**
  - **Higher Recall:** The model predicts `1` more readily, so it correctly identifies more of

the actual positive cases.
    - **Lower Precision:** The model will also incorrectly label more negative cases as positive (it will have more **False Positives, FP**).

This relationship can be visualized on a **Precision-Recall Curve**, which shows the trade-off as you vary the threshold.

The F1-Score: Combining Metrics

Since we now have two metrics, how do we compare models?

- Model A: High Precision, Low Recall
- Model B: Low Precision, High Recall

We need a single number that combines both. A simple **average** (`(P+R)/2`) is **not a good metric** because it can be "gamed" by a useless model (e.g., a model with 100% recall but 1% precision would look good).

The standard solution is the **F1-Score**.

- **What it is:** The **harmonic mean** of Precision and Recall.
- **Key Property:** The F1-Score gives more weight to the *lower* of the two values. It will only be high if **both** precision and recall are high.
- **Formula:**
  $F_1 = 2 \cdot \frac{Precision \times Recall}{Precision + Recall}$

By calculating the F1-Score, you can get a single, more reliable metric to compare different algorithms or threshold choices for your skewed dataset.